WEB PROGRAMMING

SEMESTER II

Prof. Arti Gavas

Internet



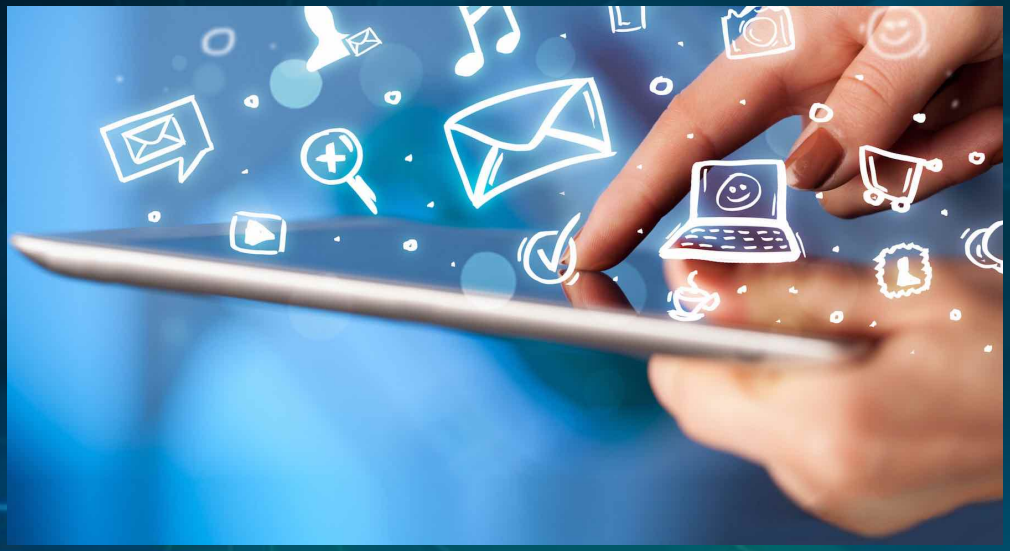
The Internet is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide.

Internet



It is a network of networks that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.

Internet



The Internet carries an extensive range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and peer-to-peer networks for file sharing.

Internet History

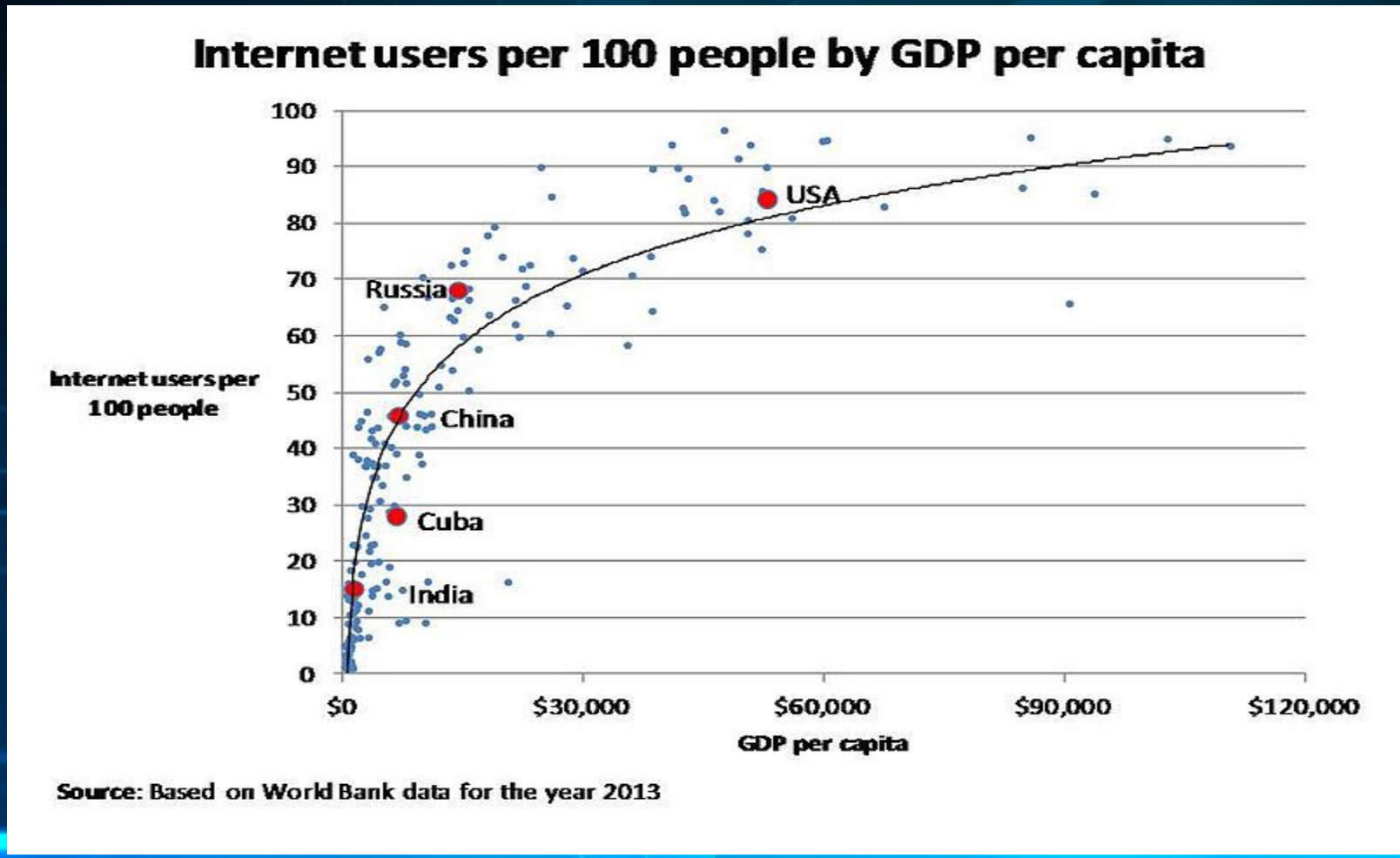
Internet use grew rapidly in the West from the mid-1990s and from the late 1990s in the developing world.

In the 20 years since 1995, Internet use has grown 100-times, measured for the period of one year, to over one third of the world population.

Most traditional communications media, including telephony, radio, television, paper mail and newspapers are being reshaped or redefined by the Internet, giving birth to new services such as email, Internet telephony, Internet television music, digital newspapers, and video streaming websites.

Newspaper, book, and other print publishing are adapting to website technology, or are reshaped into blogging, web feeds and online news aggregators.

Internet Usage



Internet Governance

The Internet has no centralized governance in either technological implementation or policies for access and usage; each constituent network sets its own policies.

Only the overreaching definitions of the two principal name spaces in the Internet, the Internet Protocol address space and the Domain Name System (DNS), are directed by a maintainer organization, the Internet Corporation for Assigned Names and Numbers (ICANN).

The technical underpinning and standardization of the core protocols is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise.

Applications

- Search engine

It can be used to search anything and everything. Most popular search engines are google and yahoo searches.



Applications

- Shopping

Shopping has become easier with the advent of internet. You can buy or sell online



Applications

- Communication

This is a major role of the internet. It helps people to communicate either with the use of social networking websites or through e mails. Even chatting is a major use of the internet.





Applications

Job search:

Nowadays, many people search for their jobs online as it is quicker and there is a larger variety of job vacancies present.

Applications

Hobbies

Those who are having certain hobbies can try to improve on it by reading up on many aspects of their hobby

Applications

Research

Research papers are present online which helps in the researcher doing a literature review.

Applications

Studying

Now right from kinder garden children are exposed to internet and computers. They find many useful things to learn on the internet(though with supervision). Up to doctorate level education, people rely on internet for their education. Online educational books have even reduced the need for a library

These are only some of the uses of the internet. There are so many more that will can be listed forever. There are also negative aspects of the internet. But the responsibility is on the user to make use of the internet for it's benefits and leave all the negative aspects.

Related Terminology

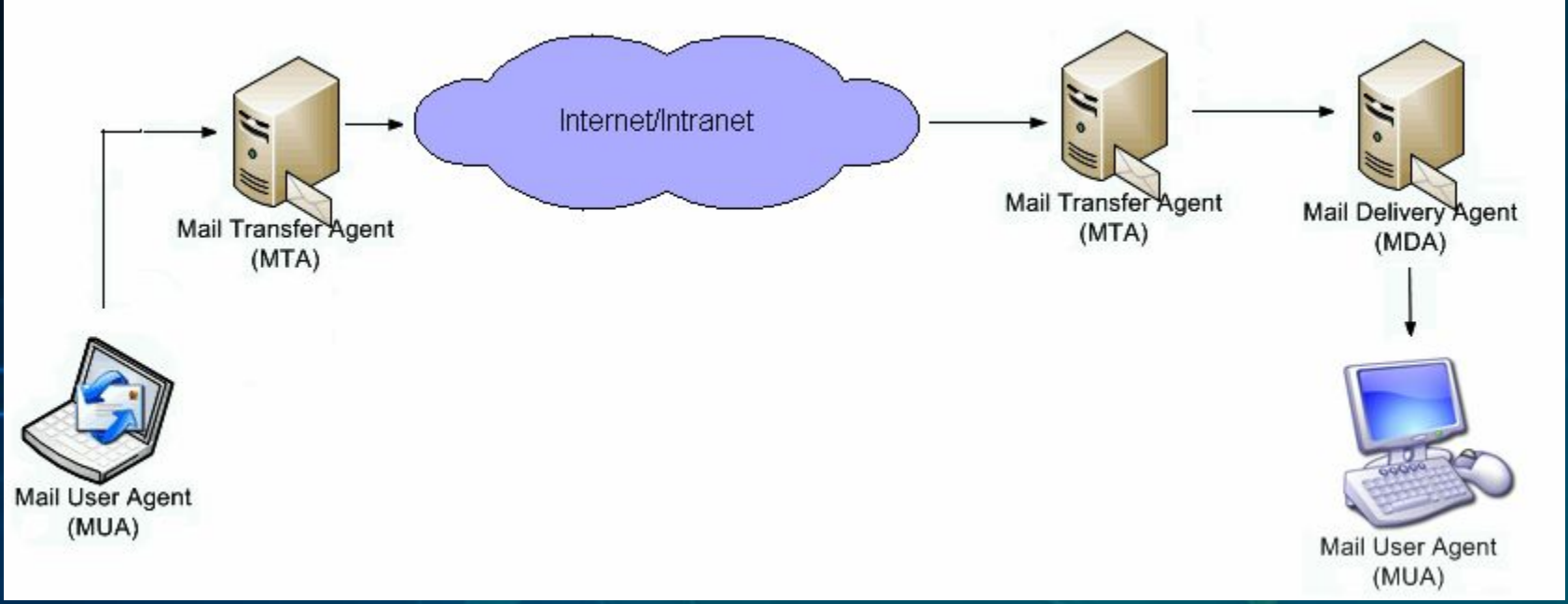
- E-mail
- telnet
- FTP
- e-commerce
- video conferencing
- e-business
- Internet service providers
- domain name server
- internet address
- World Wide Web (WWW):
- uniform resource locator (URL)
- browsers – internet explorer, Netscape navigator, opera, Firefox, chrome, Mozilla.
- search engine
- web server – apache, IIS, proxy server,
- HTTP protocol

E-mail

- Short for electronic mail,
- e-mail or email is information stored on a computer that is exchanged between two users over telecommunications.
- More plainly, e-mail is a message that may contain text, files, images, or other attachments sent through a network to a specified individual or group of individuals.
- The first e-mail was sent by Ray Tomlinson in 1971. By 1996, more electronic mail was being sent than postal mail.



E-mail


Internet/Intranet
Mail Transfer Agent
(MTA)
Mail Transfer Agent
(MTA)
Mail Delivery Agent
(MDA)
Mail User Agent
(MUA)
Mail User Agent
(MUA)

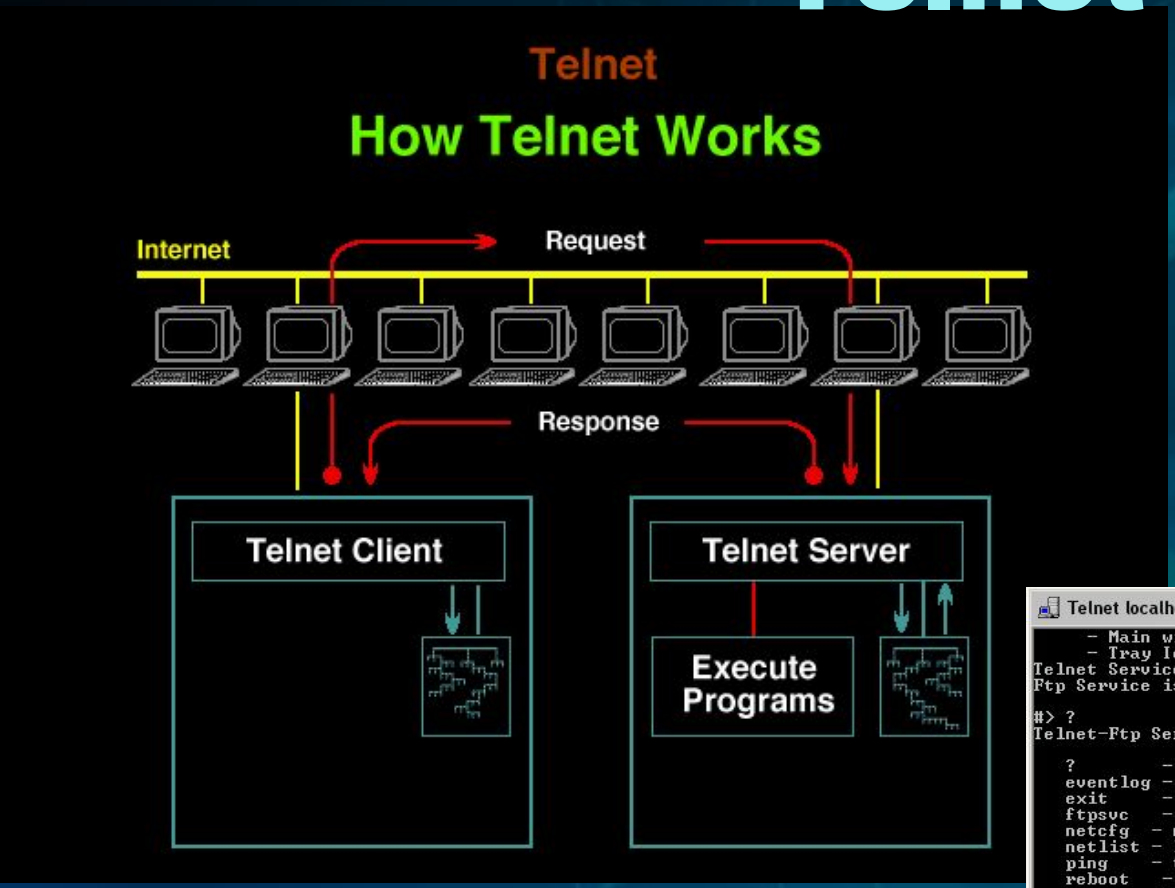
Telnet

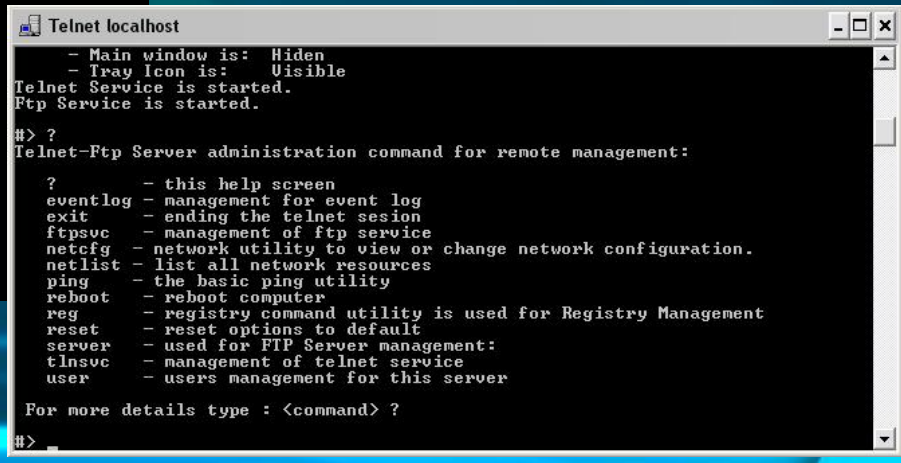
- Telnet is an application layer protocol used on the Internet or local area networks to provide a bidirectional interactive text-oriented communication facility using a virtual terminal connection.

- Telnet was developed in 1969 beginning with RFC 15, extended in RFC 854, and standardized as Internet Engineering Task Force (IETF) Internet Standard STD 8, one of the first Internet standards. The name stands for "**teletype network**".

- Historically, Telnet provided access to a command-line interface (usually, of an operating system) on a remote host, including most network equipment and operating systems with a configuration utility.

Telnet

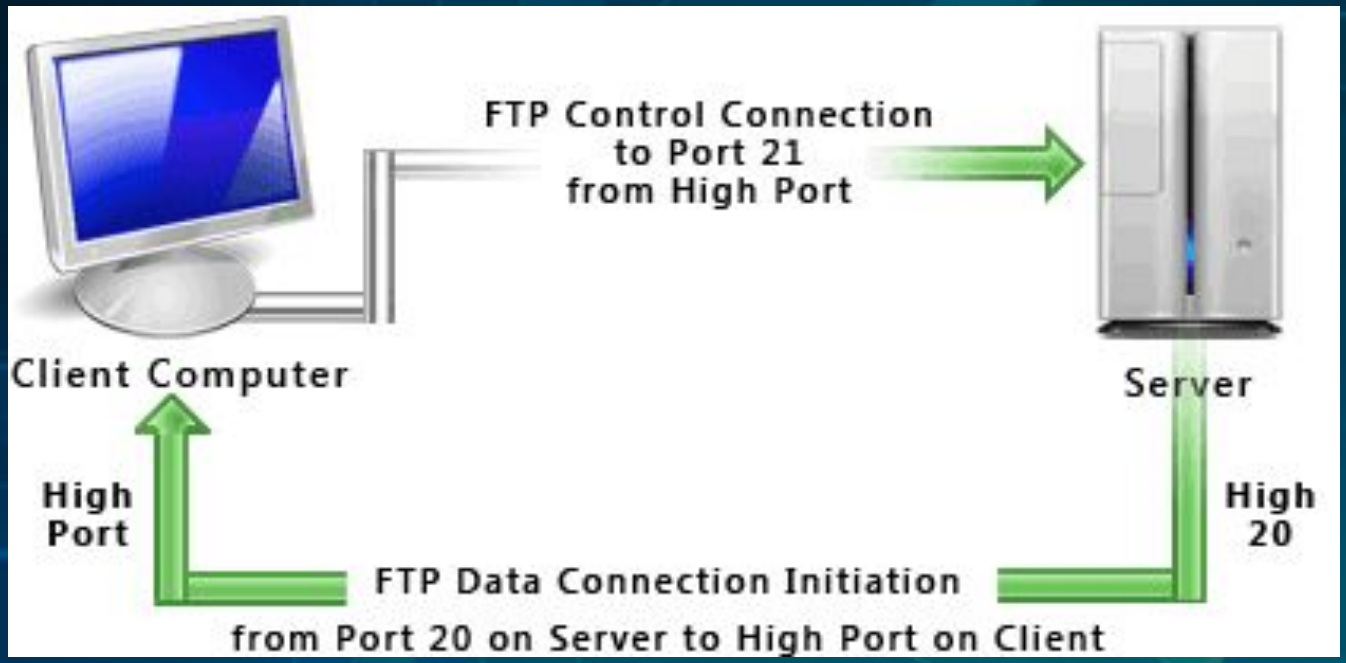

Telnet
How Telnet Works
Internet
Request
Response
Telnet Client
Telnet Server
Execute Programs


Telnet localhost
- Main window is: Hiden
- Tray Icon is: Visible
Telnet Service is started.
Ftp Service is started.
#> ?
Telnet-Ftp Server administration command for remote management:
? - this help screen
eventlog - management for event log
exit - ending the telnet sesion
ftpsvc - management of ftp service
netcfg - network utility to view or change network configuration.
netlist - list all network resources
ping - the basic ping utility
reboot - reboot computer
reg - registry command utility is used for Registry Management
reset - reset options to default
server - used for FTP Server management:
tlnsvc - management of telnet service
user - users management for this server
For more details type : <command> ?
#>

FTP

- File Transfer Protocol (FTP) is the commonly used protocol for exchanging files over the Internet.
- FTP uses the Internet's TCP/IP protocols to enable data transfer.
- FTP uses a client-server architecture, often secured with SSL/TLS.
- FTP promotes sharing of files via remote computers with reliable and efficient data transfer

FTP


FTP Control Connection
to Port 21
from High Port
Client Computer
Server
High
Port
High
20
FTP Data Connection Initiation
from Port 20 on Server to High Port on Client

e-commerce

- Electronic commerce, commonly written as e-commerce or eCommerce, is the trading or facilitation of trading in products or services using computer networks, such as the Internet or online social networks.

- Electronic commerce draws on technologies such as mobile commerce, electronic funds transfer, supply chain management, Internet marketing, online transaction processing, electronic data interchange (EDI), inventory management systems, and automated data collection systems.

- Modern electronic commerce typically uses the World Wide Web for at least one part of the transaction's life cycle although it may also use other technologies such as e-mail.

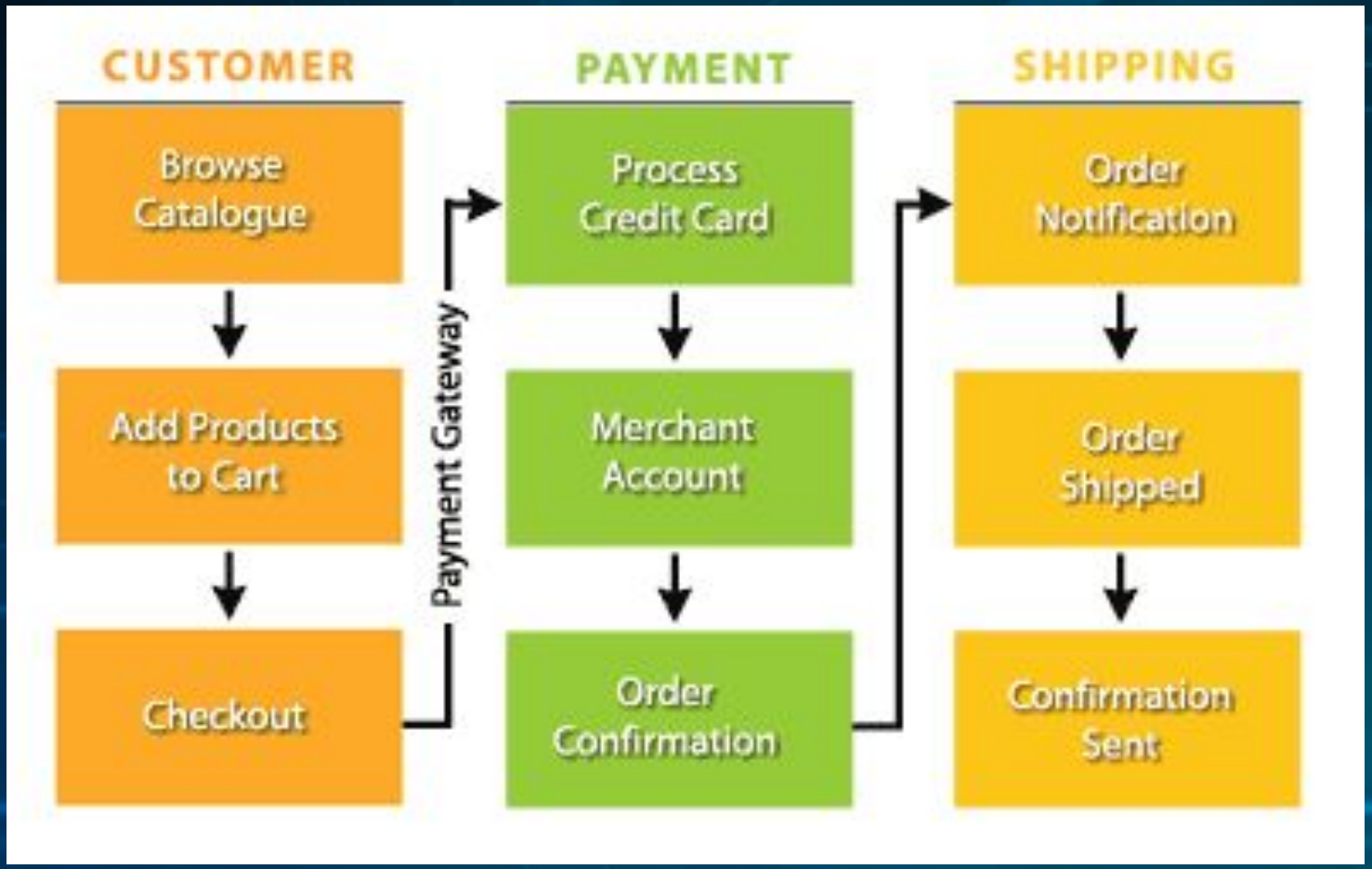


e-commerce

- E-commerce businesses may employ some or all of the following:
- Online shopping web sites for retail sales direct to consumers
- Providing or participating in online marketplaces, which process third-party business-to-consumer or consumer-to-consumer sales
- Business-to-business buying and selling
- Gathering and using demographic data through web contacts and social media
- Business-to-business (B2B) electronic data interchange
- Marketing to prospective and established customers by e-mail or fax (for example, with newsletters)
- Engaging in retail for launching new products and services
- Online financial exchanges for currency exchanges or trading purposes



e-commerce


CUSTOMER
PAYMENT
SHIPPING
Browse Catalogue
Add Products to Cart
Checkout
Payment Gateway
Process Credit Card
Merchant Account
Order Confirmation
Order Notification
Order Shipped
Confirmation Sent

video conferencing

- Videoconferencing (VC) is the conduct of a videoconference (also known as a video conference or video teleconference) by a set of telecommunication technologies which allow two or more locations to communicate by simultaneous two-way video and audio transmissions. It has also been called 'visual collaboration' and is a type of groupware.

- Videoconferencing differs from videophone calls in that it's designed to serve a conference or multiple locations rather than individuals.
- First used commercially in Germany during the late-1930s and later in the United States during the early 1970s as part of AT&T's development of Picturephone technology.
- With the introduction of relatively low cost, high capacity broadband telecommunication services in the late 1990s, coupled with powerful computing processors and video compression techniques, videoconferencing has made significant inroads in business, education, medicine and media.

video conferencing



E-business

- Electronic business, or e-business, is the application of information and communication technologies ICT in support of all the activities of business.
- Commerce constitutes the exchange of products and services between businesses, groups and individuals and can be seen as one of the essential activities of any business.
- Electronic commerce focuses on the use of ICT to enable the external activities and relationships of the business with individuals, groups and other businesses or e business refers to business with help of internet
- i.e. doing business with the help of internet network.
- The term "e-business" was coined by IBM's marketing and Internet team in 1996

E-business


Clients
Online catalogs
Sales
Auctions
Payments
Web Factory
E-Mail Factory
eBusiness
tools
My Company


Customer Management
Digitised Products Store
Online Store Setup
Global Trading
Marketing
E-BUSINESS
Order Management
Drop Shipping
Pricing Manipulations
Payment Options
Invoicing and Packaging
Shipping

Internet service providers

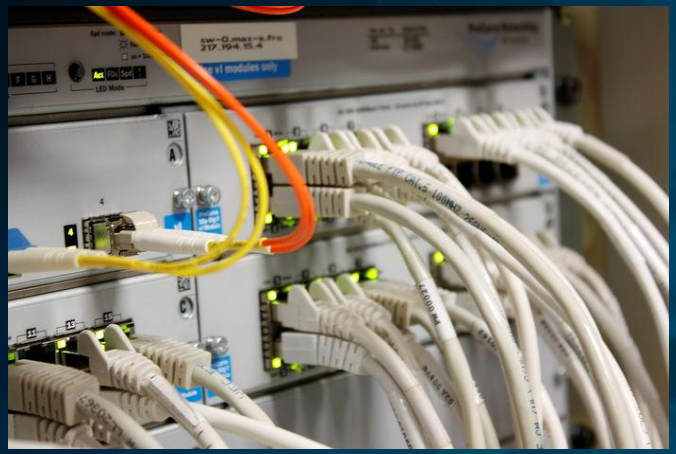
- An Internet service provider (ISP) is an organization that provides services for **accessing and using the Internet**.
- Internet service providers may be organized in various forms, such as **commercial, community-owned, non-profit, or otherwise privately owned**.
- Internet **services** typically provided by ISPs include Internet access, Internet transit, domain name registration, web hosting, Usenet service, and colocation.
- **In 1989, the first ISPs were established in Australia and the United States. In Brookline, Massachusetts, The World became the first commercial ISP in the US. Its first customer was served in November 1989.**



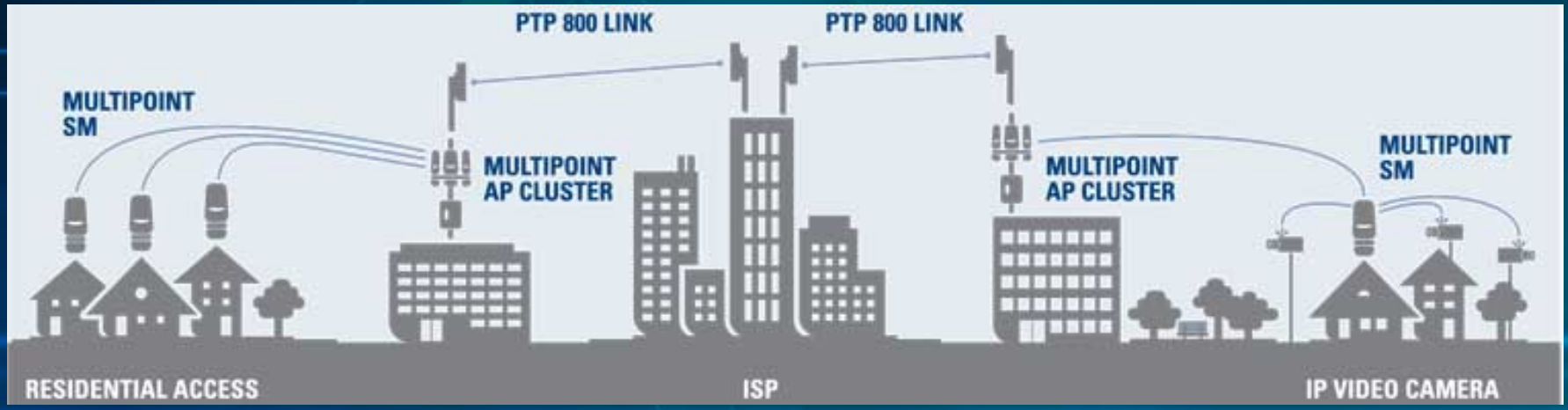
Internet service providers Classifications

- **Access providers ISP**: provide Internet access
- **Mailbox providers**: provides services for hosting electronic mail domains with access to storage for mail boxes
- **Hosting ISPs**: provide email, web-hosting, or online storage services. Other services include virtual server, cloud services
- **Transit ISPs**: transmit data to or from areas of the Internet beyond the home network: A virtual ISP (VISP) is an operation that purchases services from another ISP, sometimes called a wholesale ISP
- **Virtual ISPs**: A virtual ISP (VISP) is an operation that purchases services from another ISP, sometimes called a wholesale ISP
- **Free ISPs**: Free ISPs are Internet service providers that provide service free of charge
- **Wireless ISP**: A wireless Internet service provider (WISP) is an Internet service provider with a network based on wireless networking

Internet service providers



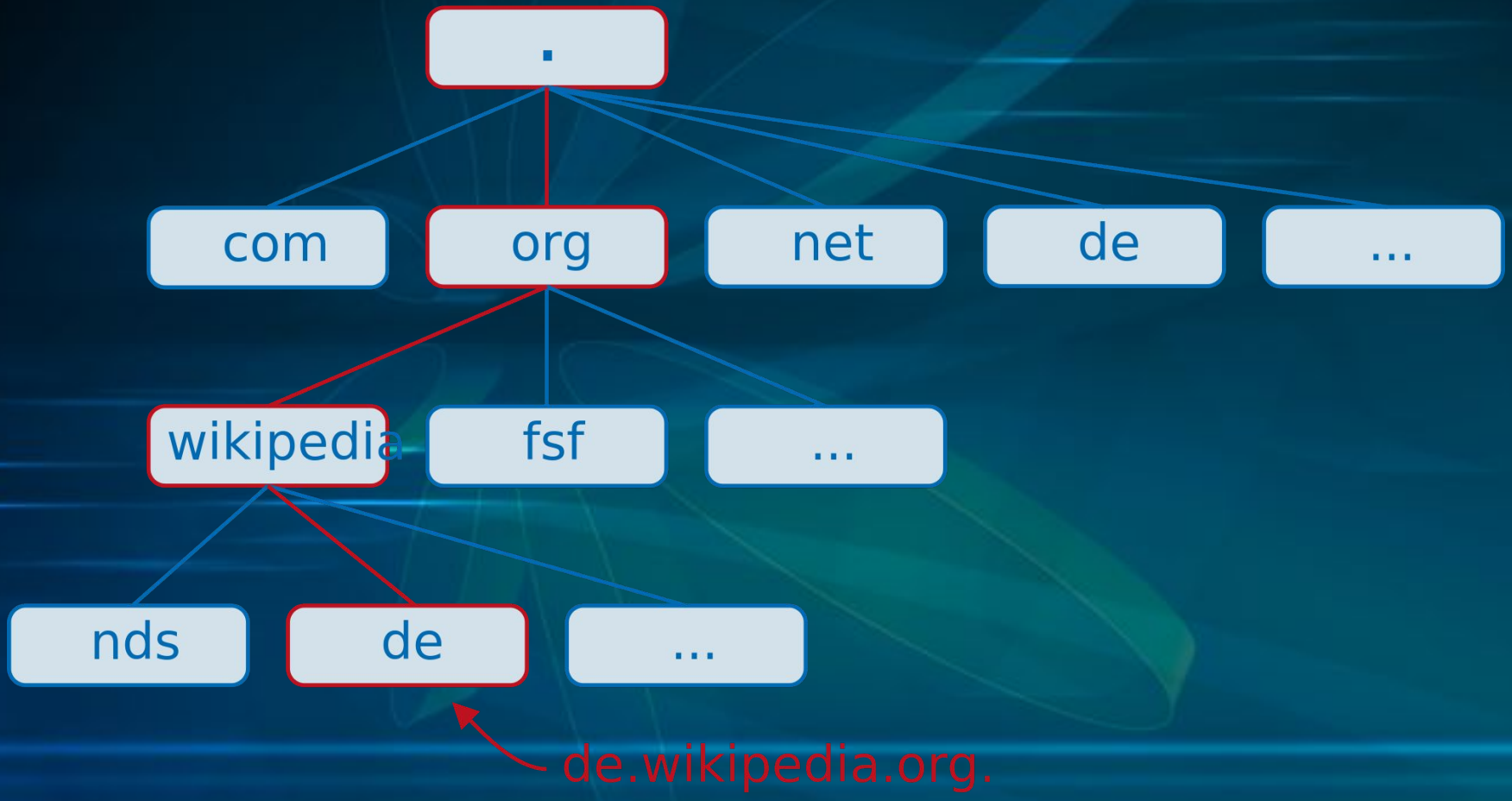




Domain Name Server (DNS)

- The Domain Name System (DNS) is a **hierarchical decentralized naming system** for computers, services, or any resource connected to the Internet or a private network.
- It associates various information with domain names assigned to each of the participating entities.
- The Domain Name System delegates the responsibility of assigning domain names and mapping those names to Internet resources by designating authoritative name servers for each domain.
- **Network administrators may delegate authority over sub-domains** of their allocated name space to other name servers.
- This mechanism provides distributed and fault tolerant service and was designed to avoid a single large central database.

DNS



internet address

- An Internet Protocol address (IP address) is a **numerical label** assigned to **each device** (e.g., computer, printer) participating in a computer network that uses the Internet Protocol for communication.

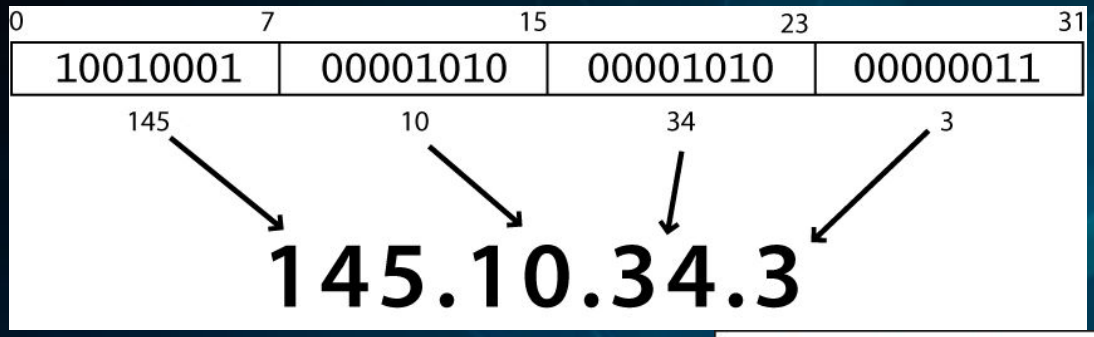
- An IP address serves two principal functions:
  - **host** or network interface **identification** and
  - **location** addressing.
- Its role has been characterized as follows:
  - "A **name** indicates what we seek.
  - An **address** indicates where it is.
  - A **route** indicates how to get there.

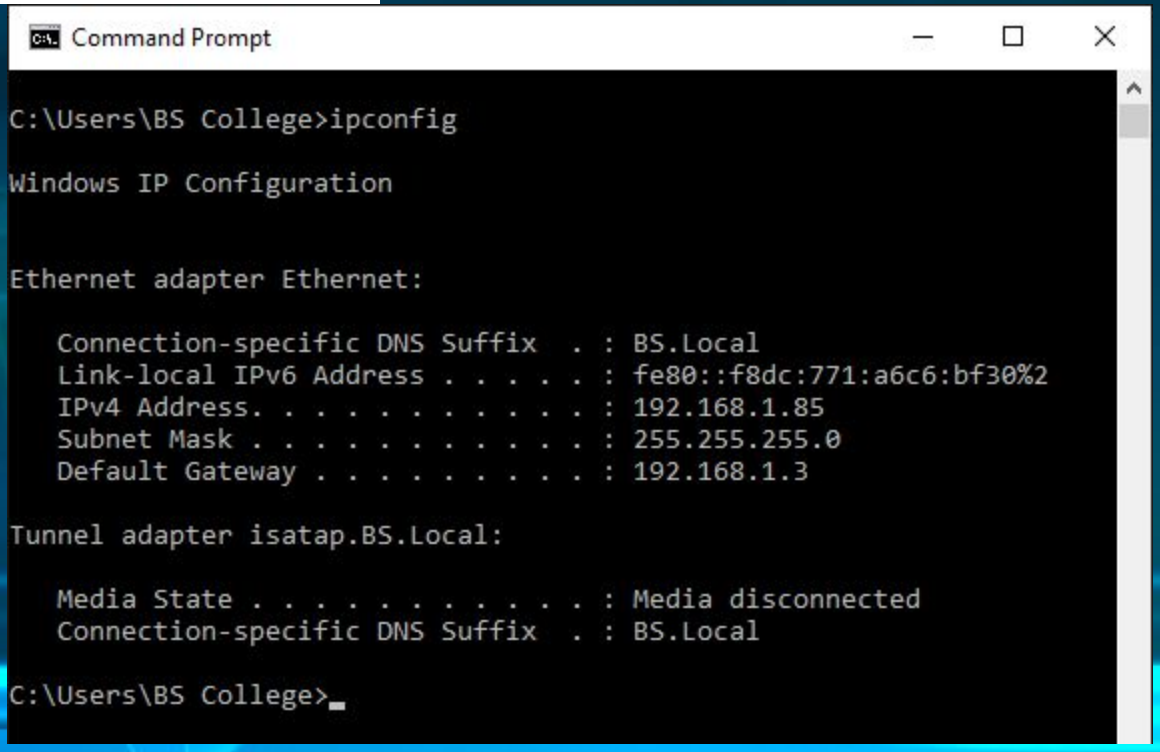
internet address

- The designers of the Internet Protocol defined an IP address as a **32-bit** number and this system, known as Internet Protocol Version 4 **(IPv4),** is still in use today.
- However, because of the **growth** of the **Internet** and the predicted depletion of available addresses, a **new version of IP (IPv6)**, using 128 bits for the address, was developed in 1995. and its deployment has been ongoing since the mid-2000s.

- IP addresses are usually written and displayed in **human-readable notations**, such as 172.16.254.1 (IPv4), and 2001:db8:0:1234:0:567:8:1 (IPv6).
- The Internet Assigned Numbers Authority (IANA) manages the IP address space allocations globally and delegates five regional Internet registries (RIRs) to allocate IP address blocks to local Internet registries (Internet service providers) and other entities.

internet address





World Wide Web (WWW)

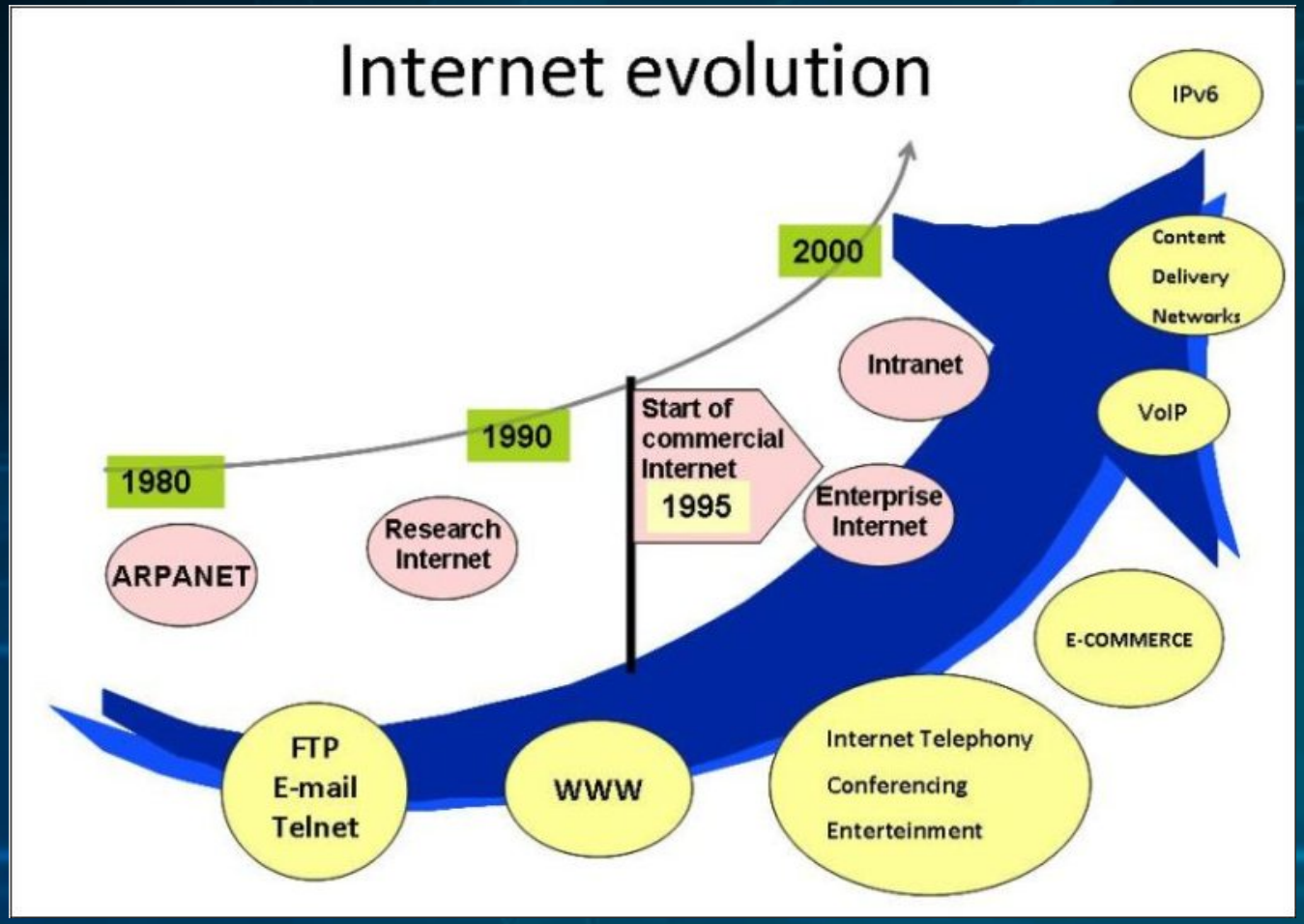
- The World Wide Web (abbreviated WWW or the Web) is an **information space** where documents and other web resources are identified by
  - Uniform Resource Locators (**URLs**),
  - interlinked by **hypertext links**, and can be accessed via the Internet.
  - English scientist Tim Berners-Lee invented the World Wide Web in 1989.
  - He wrote the first web browser computer programmed in 1990 while employed at CERN in Switzerland.
- Web pages are **primarily text documents** formatted and annotated with Hypertext Markup Language (**HTML**).
- In addition to formatted text, web pages may contain **images, video, audio**, and software components that are rendered in the user's web browser as coherent pages of **multimedia content**.
- Embedded **hyperlinks** permit users to **navigate between web pages**. Multiple web pages with a common theme, a common domain name, or both, make up a website.



WWW and its evolution

- 1980–1991: Invention and implementation of the Web
- 1992–1995: Growth of the Web
- Early browsers
- Web governance
- 1996–1998: Commercialization of the Web
- 1999–2001: "Dot-com" boom and bust
- 2002–present: The Web becomes ubiquitous

World Wide Web (WWW)



Uniform Resource Locator (URL)

- A Uniform Resource Locator (URL), commonly informally termed a **web address** is a reference to a **web resource** that specifies its **location** on a **computer network** and a mechanism for retrieving it.

- A URL implies the means to access an indicated resource. URLs occur most commonly **to reference web pages** (http), but are also used for file transfer (ftp), email (mailto), database access (JDBC), and many other applications.

- Most web browsers display the URL of a web page **above the page** in an address bar. A typical URL could have the form **http://www.example.com/index.html**,
  - which indicates a protocol (http), a hostname (www.example.com), and a file name (index.html).

Browsers

- A web browser (commonly referred to as a browser) is a software application for **retrieving, presenting, and traversing information** resources on the World Wide Web.
- An information resource is identified by a Uniform Resource Identifier (URI/URL) and may be a web page, image, video or other piece of content.
- Hyperlinks present in resources enable users easily to navigate their browsers to related resources.
- Although browsers are primarily intended to use the World Wide Web, they can also be used to access information provided by web servers in private networks or files in file systems.
- The major web browsers are Firefox, Google Chrome, Internet Explorer/Microsoft Edge, Opera, and Safari.

Browsers History

- The first web browser was invented in 1990 by **Sir Tim Berners-Lee**. Berners-Lee is the director of the World Wide Web Consortium (W3C), which oversees the Web's continued development, and is also the founder of the World Wide Web Foundation. His browser was called World Wide Web and later renamed Nexus
- The first commonly available web browser with a graphical user interface was **Erwise**.



List of Browsers

In order of release:

- WorldWideWeb, February 25, 1990
- Mosaic, April 22, 1993
- Netscape Navigator and Netscape Communicator, October 13, 1994
- Internet Explorer, August 16, 1995
- Opera, 1996, see History of the Opera web browser
- Mozilla Navigator, June 5, 2002[19]
- Safari, January 7, 2003
- Mozilla Firefox, November 9, 2004
- Google Chrome, September 2, 2008
- Microsoft Edge, July 9, 2015

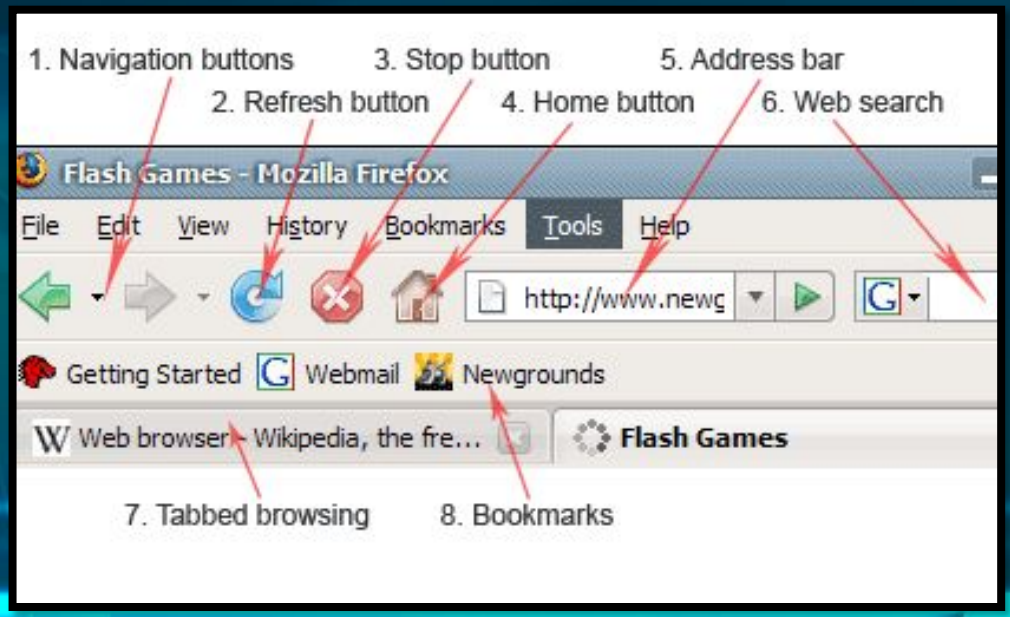


List of Browsers

Year	Web browsers	Internet users in millions
1991	WorldWideWeb (Nexus)	4
1992	ViolaWWW, Erwise, MidasWWW, MacWWW (*Samba*)	7
1993	Mosaic, Cello,[5] Lynx 2.0, Arena, AMosaic 1.0	10–14
1994	IBM WebExplorer, Netscape Navigator, SlipKnot 1.0, MacWeb, IBrowse, Agora (*Argo*), Minuet	20–25
1995	Internet Explorer 1, Netscape Navigator 2.0, OmniWeb, UdiWWW,[6] Internet Explorer 2, Grail	16–44
1996	Arachne 1.0, Internet Explorer 3.0, Netscape Navigator 3.0, Opera 2.0, PowerBrowser 1.5,[7] Cyberdog, Amaya 0.9,[8] AWeb, Voyager	36–77
1997	Internet Explorer 4.0, Netscape Navigator 4.0, Netscape Communicator 4.0, Opera 3.0,[9] Amaya 1.0[8]	70–120
1998	iCab, Mozilla	147–188
1999	Amaya 2.0,[8] Mozilla M3, Internet Explorer 5.0	248–280
2000	Konqueror, Netscape 6, Opera 4,[10] Opera 5,[11] K-Meleon 0.2, Amaya 3.0,[8] Amaya 4.0[8]	361–413
2001	Internet Explorer 6, Galeon 1.0, Opera 6,[12] Amaya 5.0[8]	499–513
2002	Netscape 7, Mozilla 1.0, Phoenix 0.1, Links 2.0, Amaya 6.0,[8] Amaya 7.0[8]	587–662
2003	Opera 7,[13] Safari 1.0, Epiphany 1.0, Amaya 8.0[8]	719–778
2004	Firefox 1.0, Netscape Browser, OmniWeb 5.0	817–910
2005	Safari 2.0, Netscape Browser 8.0, Opera 8,[14] Epiphany 1.8, Amaya 9.0,[8] AOL Explorer 1.0, Maxthon 1.0, Shiira 1.0	1018–1029
2006	SeaMonkey 1.0, K-Meleon 1.0, Galeon 2.0, Camino 1.0, Firefox 2.0, Avant 11, iCab 3, Opera 9,[15] Internet Explorer 7	1093–1157
2007	Maxthon 2.0, Netscape Navigator 9, NetSurf 1.0, Flock 1.0, Safari 3.0, Conkeror	1319–1373
2008	Konqueror 4, Safari 3.1, Opera 9.5,[16] Firefox 3, Amaya 10.0,[8] Flock 2, Chrome 1, Amaya 11.0[8]	1562–1574
2009	Internet Explorer 8, Chrome 2–3, Safari 4, Opera 10,[17] SeaMonkey 2, Camino 2, Firefox 3.5, surf	1743–1802
2010	K-Meleon 1.5.4, Firefox 3.6, Chrome 4–8, Opera 10.50,[18] Safari 5, xxxterm, Opera 11	1971–2034
2011	Chrome 9–16, Firefox 4-9, Internet Explorer 9, Maxthon 3.0, SeaMonkey 2.1–2.6, Opera 11.50, Safari 5.1	2264–2272
2012	Chrome 17–23, Firefox 10–17, Internet Explorer 10, Maxthon 4.0, SeaMonkey 2.7-2.14, Opera 12, Safari 6	2497–2511
2013	Chrome 24–31, Firefox 18–26, Internet Explorer 11, SeaMonkey 2.15-2.23, Opera 15–18, Safari 7	2712
2014	Chrome 32–39, Firefox 27–34, SeaMonkey 2.24-2.30, Opera 19–26, Safari 8	3079
2015	Microsoft Edge, Vivaldi	

Features of Browsers

- Firefox's "awesome bar"
- Safari 4's speed
- Internet Explorer 8's tabs
- Firefox's bookmark manager
- Safari 4's full-page search
- Chrome's updates
- Firefox's extendibility
- Safari's User Agent
- IE8's security features
- Chrome's most-visited

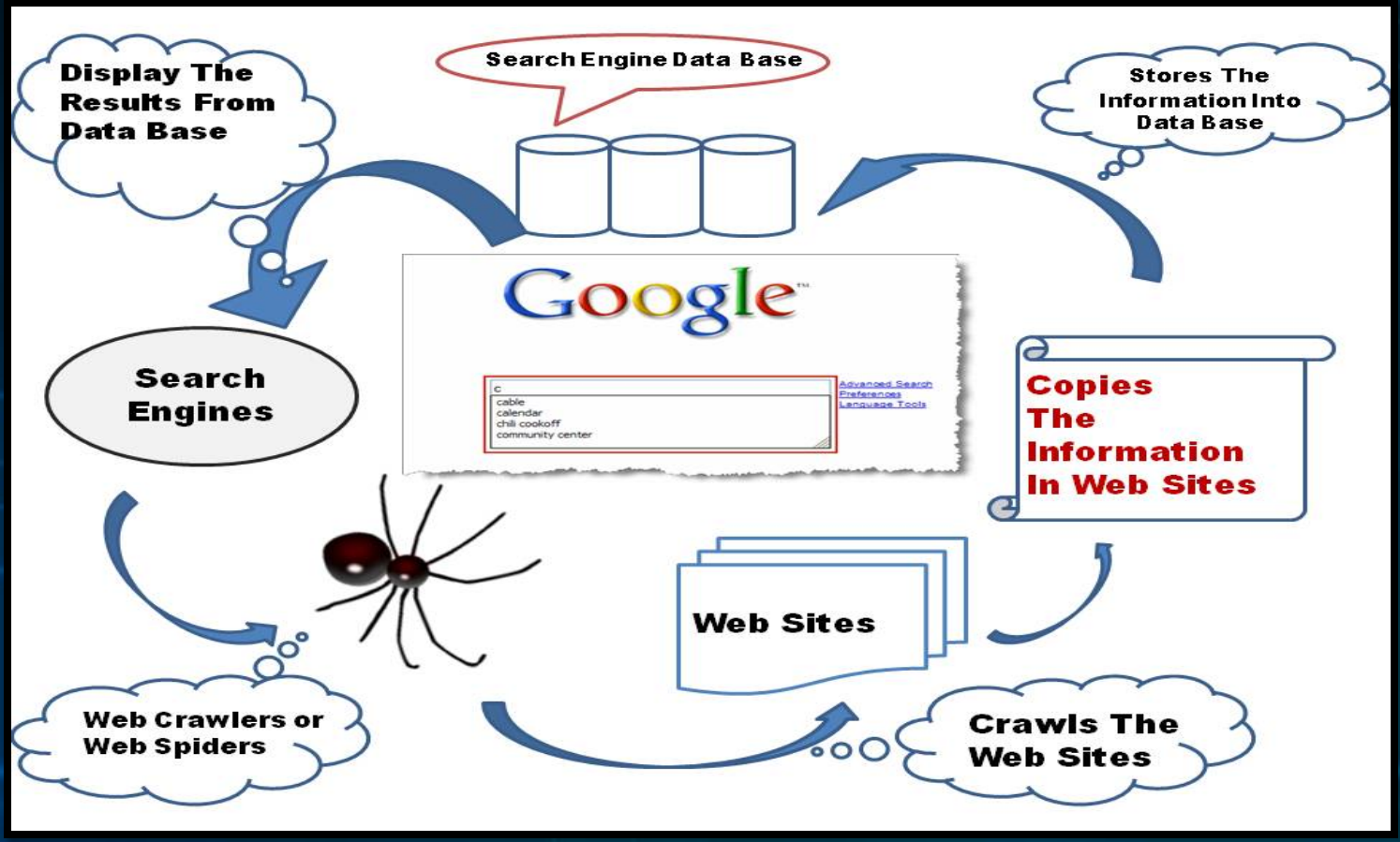


Search Engine

- A web search engine is a software system that is designed to **search for information** on the World Wide Web.

- The search results are generally presented in a line of results often referred to as search engine results pages (SERPs).
- The information may be a mix of web pages, images, and other types of files.
- Some search engines also **mine data available in databases** or open directories.
- Unlike web directories, which are maintained only by human editors, search engines also maintain real-time information by running an algorithm on a **web crawler**.

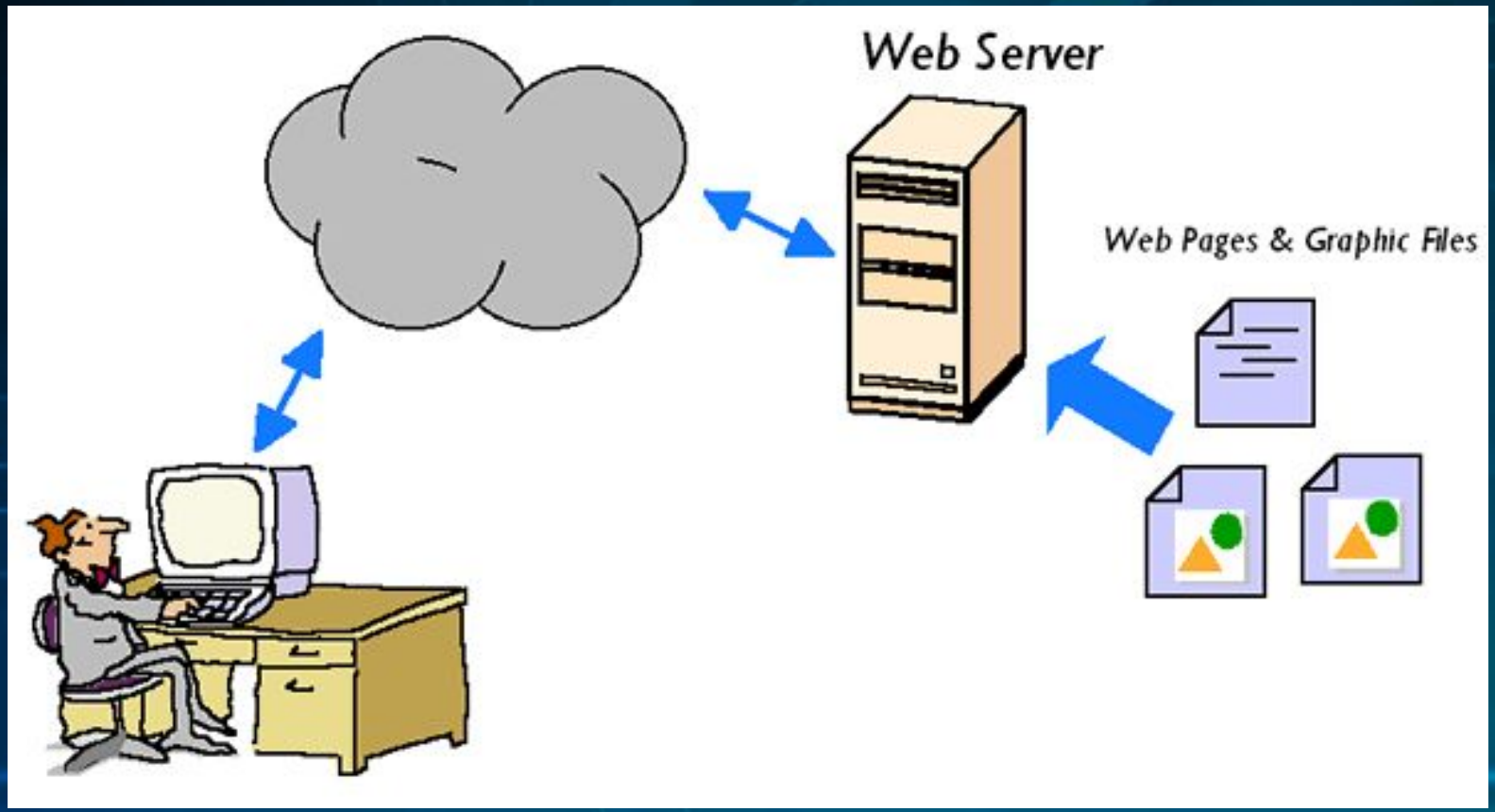
Search Engine Working



Web Servers

- A web server is a computer system that **processes requests via HTTP**, the basic network protocol used to distribute information on the World Wide Web.
- The term can refer to the entire system, or specifically to the software that accepts and supervises the HTTP requests.
- The **primary function** of a web server is to **store, process and deliver** web pages to clients.
- The communication between client and server takes place using the Hypertext Transfer Protocol (HTTP).
- Pages **delivered** are most frequently **HTML documents**, which may include images, style sheets and scripts in addition to text content.

How Web Servers work?



Different Types of Web Servers

- **Apache:** Apache is developed and maintained by an open community of developers under the auspices of the Apache Software Foundation. Most commonly used on a Unix-like system

- **IIS:** An extensible web server created by Microsoft for use with Windows NT family. IIS supports HTTP, HTTPS, FTP, FTPS, SMTP and NNTP.

- **Proxy server** : It will act as a mediator between server-to-filter requests made by users and a client program. It allows the management of emails and shared connections.
  - There are two types of proxy servers: Open Proxies and Reverse Proxies.

HTTP protocol

- The Hypertext Transfer Protocol (HTTP) is an application protocol for distributed, collaborative, hypermedia information systems.
- HTTP is the foundation of data communication for the World Wide Web.
- Hypertext is structured text that uses logical links (hyperlinks) between nodes containing text. HTTP is the protocol to exchange or transfer hypertext.
- Development of HTTP was initiated by Tim Berners-Lee at CERN in 1989.

HTML (HYPERTEXT MARKUP LANGUAGE)

HTML stands for Hyper Text Markup Language

HTML is the standard markup language for creating Web pages

HTML describes the structure of a Web page

HTML consists of a series of elements

HTML elements tell the browser how to display the content

Structure

```
<!DOCTYPE html>
<html>
<head>
   <title>.....</title>
</head>
<body>
     /* A container for all the visible contents*/
</body>
</html>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

Text Formatting Tags

1.Heading Tag

<h1> to <h6>

<h1> is the biggest and <h6> is the smallest.

Eg:

<h1>Main Heading</h1>

<h2>Sub Heading</h2>

2. Paragraph Tag

….

Eg:

This is a paragraph.

3.

Line break (no closing tag).

Eg:

First Line
Second Line

HTML (HYPERTEXT MARKUP LANGUAGE)

4. <hr>

Horizontal rule (divider line).

Eg:

```
<p>Topic 1</p>
<hr>
<p>Topic 2</p>
```

5.

Bold text (important)

Eg: Important!

6.

Italic (emphasis)

Eg: Note this

7. <mark>

Highlights text

Eg:

```
<mark>Highlight me</mark>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

8. <small>

Smaller text

Eg:<small>Fine print</small>

9.

Italic text

Eg: Sandhiya

10.

Bold text

Eg: Web Programming

11. <sup>

Superscript

Eg: 2nd

12. <sub>

Subscript

Eg: H₂O

HTML (HYPERTEXT MARKUP LANGUAGE)

MEDIA TAG

1.<img> - Images can improve the design and the appearance of a web page.

<img src="img_girl.jpg" alt="Girl in a jacket" width="500" height="600">

Attributes:

src - Specifies the path to the image

alt - Specifies an alternate text for the image

The width and height attributes always define the width and height of the image in pixels.

2. <a> - anchor tags defines a hyperlink.

link text

Attributes:

href - indicates the link's destination.

HTML (HYPERTEXT MARKUP LANGUAGE)

3. <video> - to embed video into a webpage

```
<video width="320" height="240" controls>
  <source src="https://www.w3schools.com/html/movie.mp4" type="video/mp4">
  <source src="https://www.w3schools.com/html/movie.ogg" type="video/ogg">
  Your browser does not support the video tag.
</video>
```

Attributes:

src: path to video

controls: show video controls(play, pause and volume)

width/height: set size

HTML (HYPERTEXT MARKUP LANGUAGE)

4. <audio> - to embed audio to a webpage.

```
<audio controls>
  <source src="https://www.w3schools.com/html/horse.ogg" type="audio/ogg">
  <source src="https://www.w3schools.com/html/horse.mp3" type="audio/mpeg">
  Your browser does not support the audio tag.
</audio>
```

Attributes:

src: path to audio file.

controls: Adds play/pause UI

HTML (HYPERTEXT MARKUP LANGUAGE)

LISTS

HTML lists allow web developers to group a set of related items in lists.

An unordered HTML list:

- Item
- Item
- Item
- Item

An ordered HTML list:

1. First item
2. Second item
3. Third item
4. Fourth item

HTML (HYPERTEXT MARKUP LANGUAGE)

Unordered HTML List

 - unordered tag

- List of items

NOTE: The list items will be marked with bullets (small black circles) by default:

Eg:

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Output:

An unordered HTML list

- Coffee
- Tea
- Milk

HTML (HYPERTEXT MARKUP LANGUAGE)

Unordered HTML List - Choose List Item Marker

list-style-type property is used to define the style of the list item marker. It can have one of the following values:

Value	Description
disc	**Sets the list item marker to a bullet (default)**
circle	**Sets the list item marker to a circle**
square	**Sets the list item marker to a square**
none	**The list items will not be marked**

Example:

```
<ul style="list-style-type:disc;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

HTML Ordered Lists

 - ordered list

 - List Item

Example:

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

An ordered HTML list

1. Coffee
2. Tea
3. Milk

HTML (HYPERTEXT MARKUP LANGUAGE)

Ordered HTML List - The Type Attribute

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

Example:

```
<ol type="1">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

HTML Description Lists

A description list is a list of terms, with a description of each term.

<dl> - tag defines the description list

<dt> - tag defines the term (name)

<dd> - tag describes each term

Eg:

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

A Description List

Coffee
- black hot drink

Milk
- white cold drink

HTML (HYPERTEXT MARKUP LANGUAGE)

TABLE

HTML tables allow web developers to arrange data into rows and columns.

Tags used in table

<table>

Attribute: border (line around the outside of whole table)

<caption> - to give title for displayed table

<tr> - row tag

<th> - table heading

<td> - table data

HTML (HYPERTEXT MARKUP LANGUAGE)

```
<table border="3">
<caption> NAME LIST </caption>
  <tr>
    <th>Company</th>
    <th>Contact</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Alfreds Futterkiste</td>
    <td>Maria Anders</td>
    <td>Germany</td>
  </tr>
  <tr>
    <td>Centro comercial Moctezuma</td>
    <td>Francisco Chang</td>
    <td>Mexico</td>
  </tr>
</table>
```

NAME LIST

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

HTML Table – Colspan

* To make a cell span over multiple columns, use the colspan attribute.

```
<table>
  <tr>
    <th colspan="2">Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Jill</td>
    <td>Smith</td>
    <td>43</td>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>57</td>
  </tr>
</table>
```

Name		Age
Jill	Smith	43
Eve	Jackson	57

HTML Table - Rowspan

To make a cell span over, multiple rows, use the rowspan attribute.

```
<table>
  <tr>
    <th>Name</th>
    <td>Jill</td>
  </tr>
  <tr>
    <th rowspan="2">Phone</th>
    <td>555-1234</td>
  </tr>
  <tr>
    <td>555-8745</td>
</tr>
</table>
```

Name	Jill
Phone	555-1234
	555-8745

HTML Table - Rowspan

To make a cell span over, multiple rows, use the rowspan attribute.

```
<table>
  <tr>
    <th>Name</th>
    <td>Jill</td>
  </tr>
  <tr>
    <th rowspan="2">Phone</th>
    <td>555-1234</td>
  </tr>
  <tr>
    <td>555-8745</td>
</tr>
</table>
```

Name	Jill
Phone	555-1234
	555-8745

HTML (HYPERTEXT MARKUP LANGUAGE)

FORMS

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

* The <form> Element

* The HTML <form> element is used to create an html form for user input.

- The <form> element is a container for different types of input elements, such as: text fields, checkboxes, radio button, submit buttons:

The <input> Element

An <input> element can be displayed in many ways , depending on the type attribute.

Type	Description
<input type="text">	Displays a single-line text input field
<input type="radio">	Displays a radio button (for selecting one of many choices)
<input type="checkbox">	Displays a checkbox (for selecting zero or more of many choices)
<input type="submit">	Displays a submit button (for submitting the form)
<input type="button">	Displays a clickable button

HTML (HYPERTEXT MARKUP LANGUAGE)

FORMS

Text Box

Allows users to input a single line of text

Radio Button

Allows selection of one option from a group

Checkbox

Allows selection of multiple options

Text Area

Allows multi-line text input

Select (Dropdown menu)

Allows selection from a dropdown list.

File Upload

Allows user to upload files.

Button

submits the form or Tiggers actions.

HTML (HYPERTEXT MARKUP LANGUAGE)

FORMS

Text Fields

The <input type=“text”> defines a single line input field for text input.

```
<form>
 Name:  <input type="text" name="fname“ value=“Enter Name” size=“15” maxlength=“20”><br>
Password: <input type=“password” name=“password”>
```

Radio Buttons

The <input type=“radio”> defines a radio button.

```
 <input type="radio" name=“gender" checked=“checked”>Male<br>
 <input type="radio" name=“gender" >Female
```

HTML (HYPERTEXT MARKUP LANGUAGE)

FORMS

Checkboxes

```
<input type="checkbox" name="hobbies" >Singing <br>
<input type="checkbox" name="hobbies"> Playing
```

Text Area

```
<textarea rows="5" cols="20">
</textarea>
```

Select

```
<select size="3" multiple>
<option name="course" selected> C</option>
<option> C++ </option>
<option> Java </option>
<option> Fullstack Developer</option>
</select>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

FORMS

FileUpload:

<input type=“file” name=“file” accept=“ “>

Output

Choose File no files choosen

The Submit Button

<input type="submit" name=“button” value="Submit">

<input type=“reset” name=“button” value=“Reset”>

<input type=“Button” name=“button” value=“click”>

<input type=“image” name=“button” src=“path”>

HTML (HYPERTEXT MARKUP LANGUAGE)

Semantic tags

<article> Represents independent, self-contained content

<aside> Contains content tangentially related to the main content, like a sidebar.

<footer> Represents the footer of the document, containing information about the author, related links, etc..

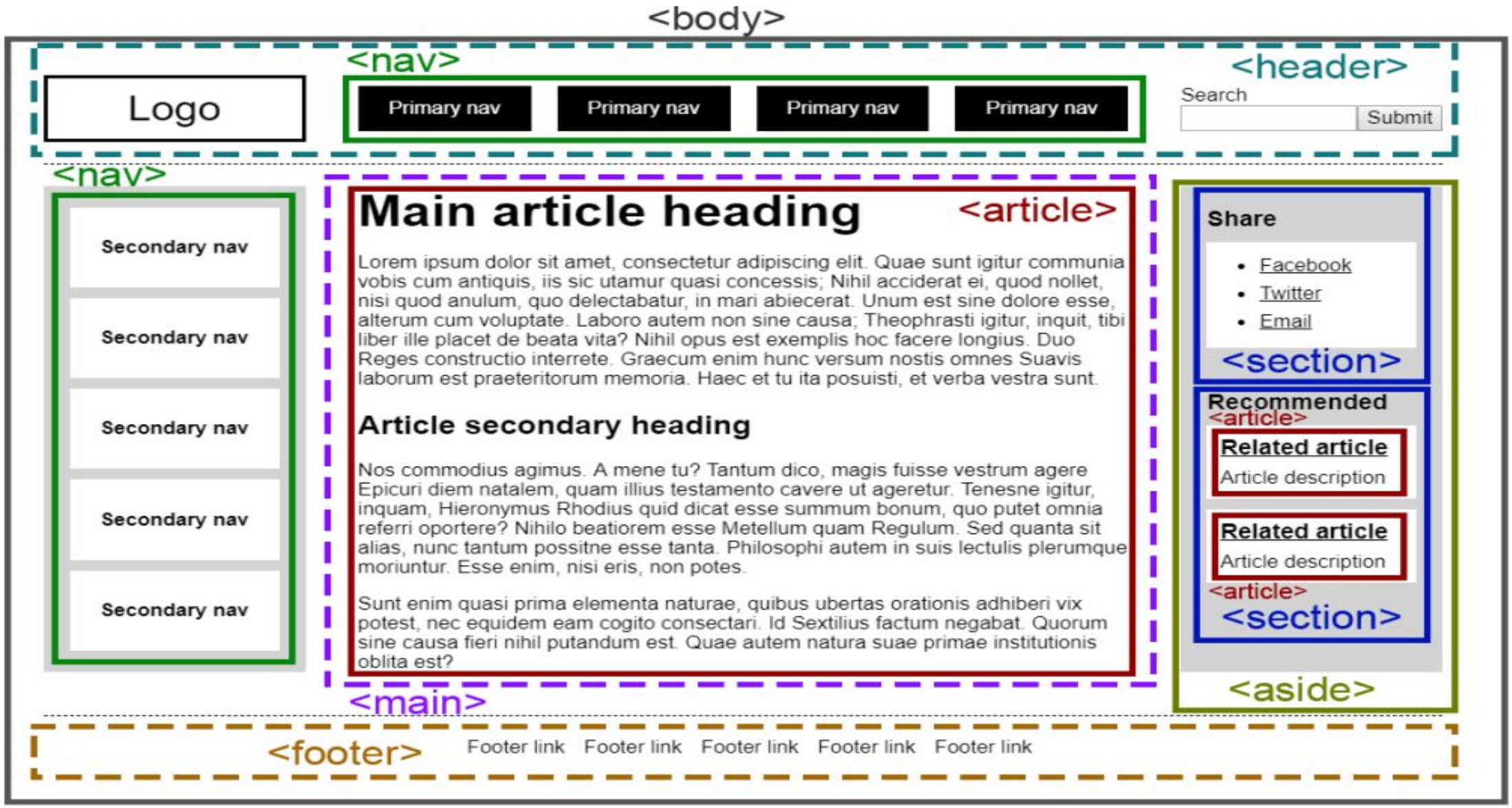
<header> Defines the introductory content or header of a page.

<main> Specifies the main content of a document.

<nav> Designates navigation links.

<section> Designates thematically connected generic content.

HTML (HYPERTEXT MARKUP LANGUAGE)

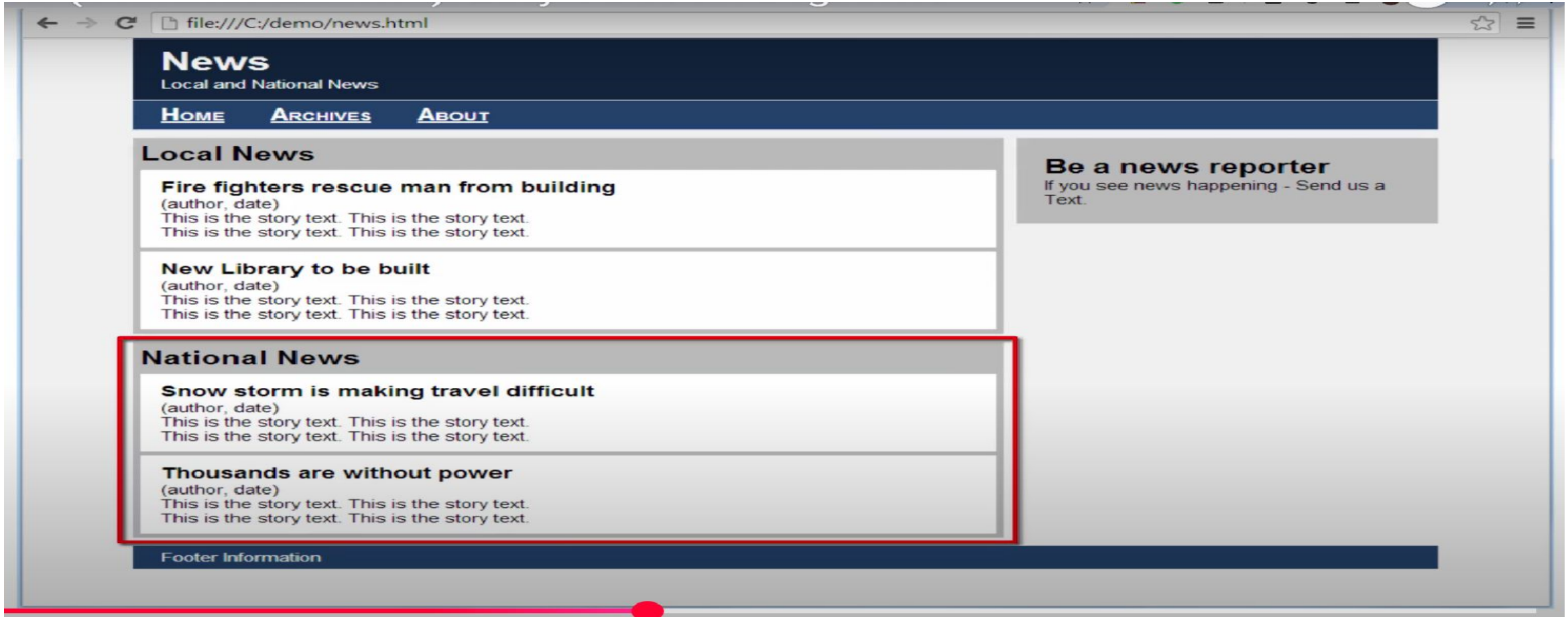


HTML (HYPERTEXT MARKUP LANGUAGE)

SECTION TAG

The HTML < SECTION > tag is a semantic HTML tag used to define a section in the document that generally includes a group of related content.

We use section tag to organize complex documents.



HTML (HYPERTEXT MARKUP LANGUAGE)

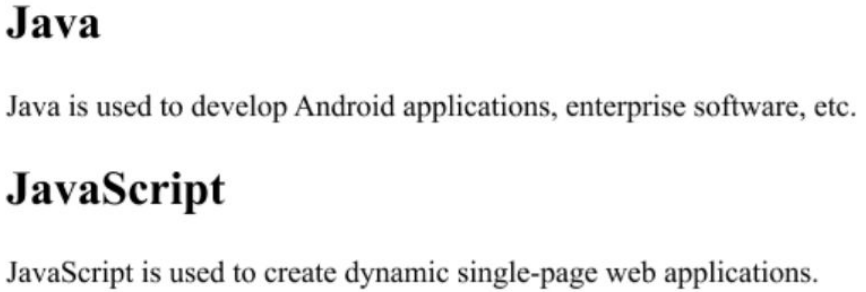
SECTION TAG

Example:

```
<section>
   <h2>Java</h2>
   <p>Java is used to develop Android applications, enterprise software, etc.</p>
</section>

<section>
   <h2>JavaScript</h2>
   <p>JavaScript is used to create dynamic single-page web applications.</p>
</section>
```

Note: always having a heading.



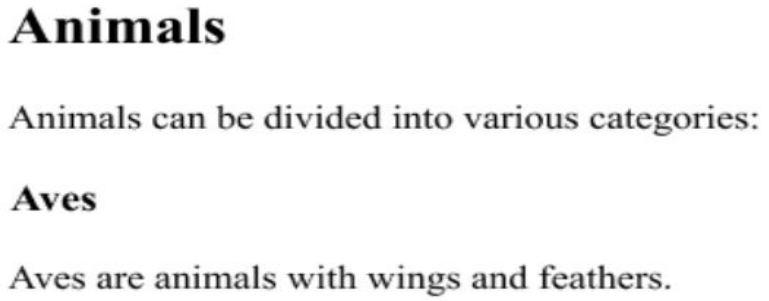
HTML (HYPERTEXT MARKUP LANGUAGE)

SECTION TAG

Nested <section> tag

<section> inside a <section> tag. It is called a nested <section> tag.

```
<section>
   <h2>Animals</h2>
   <p>Animals can be divided into various categories: </p>
   <section>
      <h3>Aves</h3>
      <p>Aves are animals with wings and feathers. </p>
   </section>
</section>
```



HTML (HYPERTEXT MARKUP LANGUAGE)

ASIDE TAG

The HTML <aside> tag is used to represent a portion of a document that is indirectly related to the main content. It is most commonly used as a sidebar in the document.

```
<div>
   <h1>This is a heading</h1>
   <aside>
      <p>This is a paragraph</p>
   </aside>
</div>
```



Here , the <aside> element contains paragraph that is indirectly related to the <h1> element.

HTML (HYPERTEXT MARKUP LANGUAGE)

ARTICLE TAG

The <article> tag specifies independent, self-contained content.

An article should make sense on its own and it should be possible to distribute it independently from the rest of the site.

Potential sources for the <article> elements:

Forum post

Blog Post

News story

Example

```
<article>
  <h2>5 Tips for Writing Better Code</h2>
  <p>Writing clean and maintainable code requires discipline and good habits...</p>
</article>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>News Example with Semantic Tags</title>
</head>
<body>
<h1>Daily News Portal</h1>
  <!-- Article Tag Example -->
  <article>
    <h2>Article: Floods Hit Southern Region</h2>
    <img src="https://via.placeholder.com/300x150?text=Flood+News" alt="Flood image">
    <p>Heavy rains have caused severe flooding in the southern districts, displacing thousands of families.</p>
    <p>Rescue operations are ongoing and temporary shelters have been set up for affected people.</p>
  </article>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

```
<!-- Section Tag Example -->
 <section>
  <h2>Section: Sports Headlines</h2>
  <h3>Cricket: India Wins Series</h3>
  <img src="https://via.placeholder.com/300x150?text=Cricket+Victory" alt="Cricket image">
  <p>The national team secured a 2-1 victory against Australia in a thrilling final match.</p>
  <h3>Tennis: Star Player Advances</h3>
  <img src="https://via.placeholder.com/300x150?text=Tennis+Match" alt="Tennis image">
  <p>The Indian tennis star has reached the semifinals in the Singapore Open.</p>
 </section>

 <!-- Aside Tag Example -->
 <aside>
  <h2>Aside: Weather Alert</h2>
  <img src="https://via.placeholder.com/300x150?text=Weather+Alert" alt="Weather image">
  <p>Heavy rainfall expected in coastal areas today. Avoid unnecessary travel.</p>
  <p>Related: <a href="#">How to stay safe during floods</a></p>
 </aside>
</body>
</html>
```

CSS (CASCADING STYLE SHEET)

CSS stands for Cascading Style Sheets

CSS describes how HTML elements are to be displayed on screen, paper, or in other media

CSS saves a lot of work. It can control the layout of multiple web pages all at once

External stylesheets are stored in CSS files

EXAMPLE

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: lightblue;
}

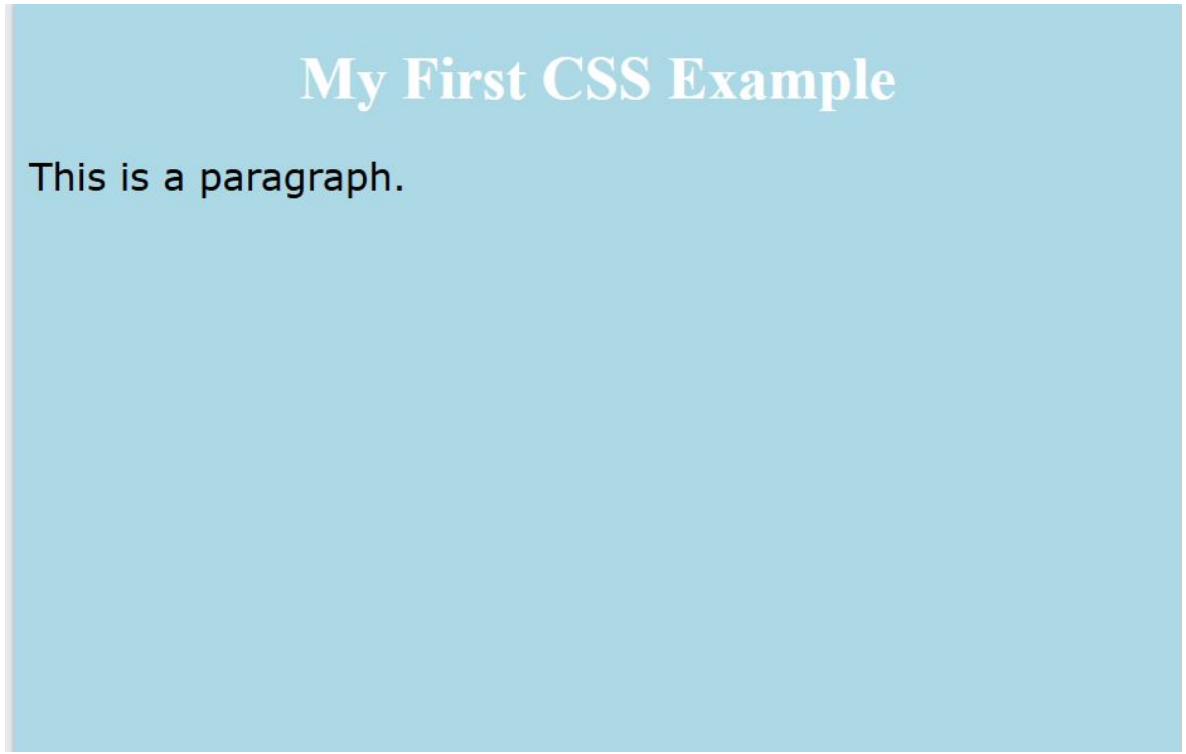
h1 {
  color: white;
  text-align: center;
}
```

HTML (HYPERTEXT MARKUP LANGUAGE)

```
p {
  font-family: verdana;
  font-size: 20px;
}
</style>
</head>
<body>

<h1>My First CSS Example</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

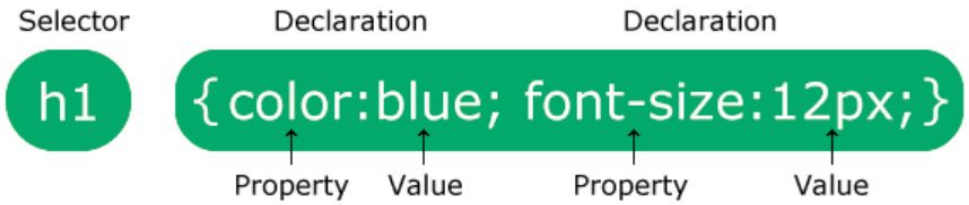


HTML (HYPERTEXT MARKUP LANGUAGE)

CSS CAN BE ADDED IN 3 WAYS

- Inline
- Embedded or Internal
- External

General syntax of CSS



- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.

HTML (HYPERTEXT MARKUP LANGUAGE)

INLINE

- Used to specify the style to a single HTML element
- The attribute used is <style>

Syntax

<elementname style =“property name:property value;”>

content

<elementname>

Example:

<h1 style=“color: blue”>Hai</h1>

HTML (HYPERTEXT MARKUP LANGUAGE)

Inline:

```
<html>

<body>

<p style="color: red">Web technology encompasses the tools, protocols, and programming languages used to build, deliver, and interact with websites and web applications. </p>

<h1 style="text-align:center; color:blue; font-size:20">WELCOME TO TH ECOURSE</h1>

<h1>full stack developer</h1>

</body>

</html>
```

Output

Web technology encompasses the tools, protocols, and programming languages used to build, deliver, and interact with websites and web applications.

WELCOME TO TH ECOURSE

full stack dveloper

Internal Stylesheet

- Used to define for a single html page
- <Style> tag is used in <head> tah

Syntax:

```
< style type="text/css">
Tagname
{
    Property name: property value;
}
</style>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

Ex:

```
<!DOCTYPE html>
<html>

<head>
    <title>Internal CSS</title>
    <style type="text/css">
        h1 {
            color: green;
            font-size: 50px;
            text-align: center;
        }

        p {
            color: blue;
            font-size: 25px;
            line-height: 1.5;
            text-align: center;
        }
    </style>
</head>

<body>
    <h1>GeeksforGeeks</h1>
    <p>A Computer Science Portal..!</p>
    <p> Web programming</p>
</body>

</html>
```



HTML (HYPERTEXT MARKUP LANGUAGE)

External Style sheet:

- Used to define style pf many html pages.
- Separate CSS file will be created and saved with the extension .css

Syntax:

<link rel="stylesheet" type="text/css" href="filename.css">

*<link> tag is used within <head> tag.

*rel attribute provides relation between css and html file.

Eg:

```
<html>
<head>
<link rel="stylesheet" href="styles.css" type="text/css">
</head>
<body>
<h1> hai</h1>
</body>
</html>
```

HTML (HYPERTEXT MARKUP LANGUAGE)

Style.css

```
body
{
Background-color:blue;
}
h1
{
Color:red
}
```

DIV TAG(Generic container for **any content)**

- **Block-level** element
- Groups **large sections** of content
- Starts on a **new line** and takes full width
- Used to create sections like header, footer, sidebar.

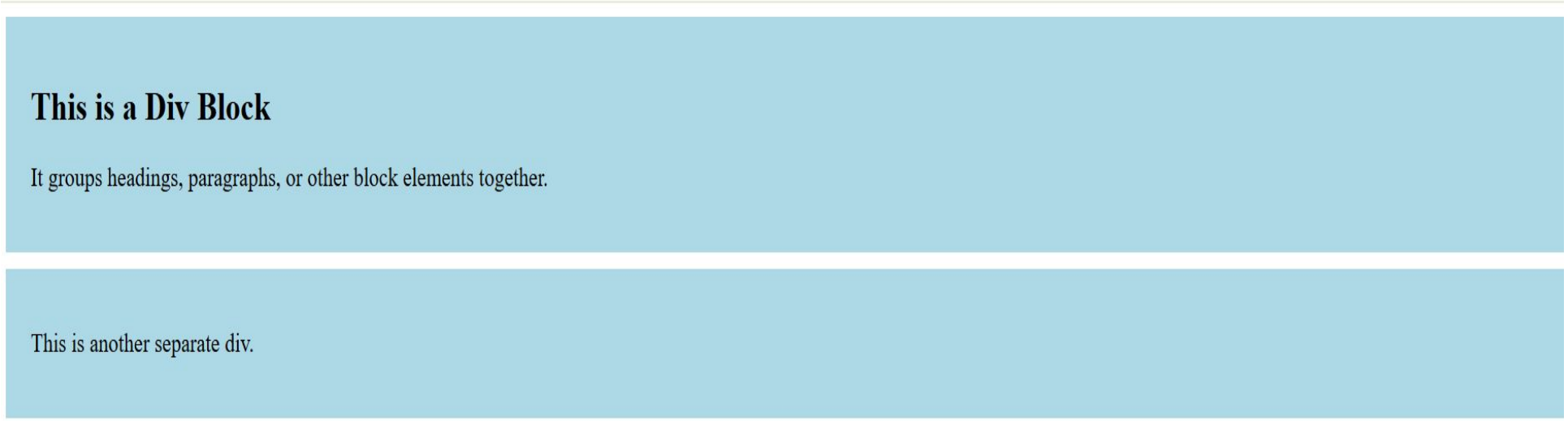
```
<!DOCTYPE html>
<html>
<head>
  <style>
    div {
      background-color: lightblue;
      padding: 20px;
      margin-bottom: 10px;
    }
  </style>
</head>
<body>
```

DIV TAG

```
<div>
  <h2>This is a Div Block</h2>
  <p>It groups headings, paragraphs, or other block elements together.</p>
</div>

<div>
  <p>This is another separate div.</p>
</div>

</body>
</html>
```



DIV TAG

Example2:

```
<!DOCTYPE html>
<html>
<head>
  <title>Web Programming</title>
</head>
<body>


  <div style="background-color: red; height: 100px; border: 2px dotted green; padding: 15px; color: white;">
    <p>Sandhiya</p>
    <h1>Welcome to Web Programming</h1>
  </div>


</body>
</html>
```

Span TAG

- **Inline** element
- Groups **small parts** within a line
- Stays **within the line**, only takes up as much space as needed

```
<html>
<head>
  <style>
    span {
      color: red;
      font-weight: bold;
    }
  </style>
</head>
<body>

  <p>This is a sentence with a <span>red and bold word</span> using span tag.</p>
  <p>Another example with <span>highlighted text</span> in the middle.</p>

</body>
</html>
```



Types of CSS Selectors

1. CSS Simple Selector or Element selector
2. CSS Id Selector
3. CSS class Selector
4. CSS Universal Selector
5. CSS Group selector
6. CSS Pseudo classes

CSS Simple selector (or) Element selector

* property values in the rule apply to all occurrences of named element.
* selects the element by name

eg:
```
<style>
P {text-align:center;
   color:blue;
  }
</style>
<body> <p> Hai </p> </body>
```

eg:
```
h1 {font-size:24pt;}
h2,h3 {font-size:20pt;}
```

CSS Id selector

* selects the id attribute of an html element to select a specific element

Syntax:
```
#specificid { property-value list }
```

```
<style>
 #para1 {
        ....
        }
</style>
<p id="para1"> Hai </p>
```

Scanned by CamScanner

CSS class selector

*select HTML elements with a specific class attribut

eg:
```
<style>
 .center
  {....
  }
 <body>
 <h1 class="center"> WebTech </h1>
 <p class="center"> ..... </p>
```

class selector for specific element

```
<style>
P.center
  {....
  }
</style>
<body>
 <h1 class="center"> Web Tech </h1>
 <p class="center"> .... </p>
```

Universal selector

*selects all elements of pages

eg:
```
<style>
 * { color:green;
   }
 </style>
 <body>
  <h2> .... </h2>
  <p> ..... </p>
  </body>
```

Scanned by CamScanner

Group selector

*to select all the elements with same style definitions.
*minimize code
*commas are used.

eg:
```
h1,h2,p
{
  text-align:center;
  color:blue;
}
```

Pseudo-class selector

*applies on associated element when mouse cursor is over it

hover :- when the cursor moves over the element
focus :- an element acquires focus when the user places the mouse cursor over it and clicks left mouse button.

eg:
```
<style>
input:hover{color:red;}
input:focus {color:green;}
</style>
<body>
<form>
Name <input type="text"/>
</form>
</body>
```

Background Properties

*used to define the background effects of an element.

i) background-color → color of background
ii) background-image → set background images for an element.

eg:
```
h1
{
  background-image:url("paper.gif");
}
```

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iii) Background-position → how to position a background image
→ used to specify the position of background image

syntax:
background-position : value ;

left top	center top	right top
left center	center center	right center
left bottom	center bottom	right bottom

iv) Background-repeat → used to repeat image horizontally or vertically
* repeat-x :- horizontally
* repeat-y :- vertically
* no-repeat (default
* repeat-xy

v) background-attachment :- sets whether a background image scrolls with the rest of page or is fixed.

syntax:
background-attachment : value
↓
fixed (or) scroll

eg:
```
body
  {
   background-image: url("tree.jpg");
background-image repeat: nonepeat;
   background-position: right top;
   background-attachment: fixed;
  }
```

Scanned by CamScanner

Text Properties

→ used to format and style text

1) color :- used to set the color of text

syntax : color : value ;

eg: color : red;

2) text-align :- used to set horizontal alignment of text.

syntax : text-align : value;

left right center justify

3) text-decoration :- used to set or remove decorations from text.

syntax : text-decoration : value;

overline underline line-through none [remove underline from link]

4) text-transform :- used to perform transformation on text.

syntax : text-transform : value ;

uppercase lowercase capitalize [first letter of each word]

5) text-indent :- used to specify the identation of the first line of text.

syntax : text-indent : value ; eg) text-indent : 20 px ;

6) letter-spacing :- used to specify the space between the characters in text.

syntax :- letter-spacing : value

eg) 3px increase -3px decrease

7) Word-spacing :- used to specify the space between the words in a text.

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8) line-height: used to specify the space between lines.

9) direction: to change the text direction of an element.

syntax: direction: value;

ltr rtl

10) text-shadow: adds shadow to the text

syntax: text-shadow: horizontal shadow vertical shadow color;

eg: text-shadow: 3px 3px red;

Font Properties

* Defines font family, boldness, size and style of a text.

1) font-family: used to set font type of an html element

syntax font-family: value;

eg: font-family: "Times New Roman";

2) font-style: used to specify style of text

values: normal, italic, oblique

3) font-size: * used to specify size of text
* default size is 16px (1em)

4) font-weight: to specify weight of font

values → normal
→ bold

5) font-variant: specifies whether or not a text should be displayed in small-caps font

values: → normal
→ small-caps

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3) tag:

to apply font properties to a word or phrase in a line.

eg: Welcome to Web Tech

```
<style>
  .s1 {
      font-size: 24pt;
      font-family: TimesNewRoman;
      color: blue;
      }
```

 Tag:- (division tag)

to apply a style to a section of document rather than each paragraph.

eg:
```
<body>
  <p>. . . . . </p>
  <div style="background-color:blue">
  <h1> . . . . . </h1>
  <p> . . . . . </p>
  </div>
  <p> . . . . . </p>
  </body>
```

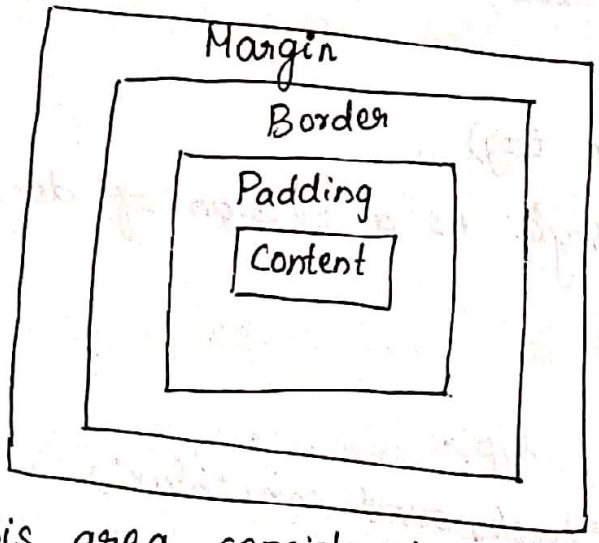
BOX MODEL

*It is a container which contains multiple properties including

i) border
ii) Margin
iii) padding
iv) content

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* used to create the design and layout of web pages
* web browser renders every element as a rectangular box according to css box model.
* used as a tool kit for customizing the layout of different elements.



Content:- This area consists of content like text, image, media etc...

Padding:- space around the content of an element and its border

Border:- area between box's padding and margin.

Margin:- space between border and an adjacent element.

Border Properties

* border-style
* border-width
* border-color

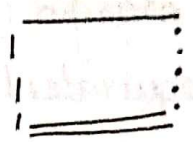
border-style:- used to specify what kind of border to display

values i) dotted
ii) dashed
iii) solide
iv) double
v) none

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The styles of one of the four sides of border can be specified as

i) border-top-style : solid;
ii) border-bottom-style : double;
iii) border-left-style : dashed;
iv) border-right-style : dotted;



The styles of all the four sides of a border can be specified with shorthand property as

border-style : solid; → 

border-width: * used to specify thickness of border.
* values in (px, pt, cm, em) or (thin, medium, thick)

The width of one of the four sides of border can be specified as

i) border-top-width
ii) border-bottom-width
iii) border-left-width
iv) border-right-width

The shorthand property can be specified as

border-width : 10px;

border-color :- to set color of the borders
* individual borders of an element can be colored as

i) border-top-color
ii) border-bottom-color
iii) border-left-color
iv) border-right-color

The shorthand property can be specified as:

border-color : red;

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Margin properties

```
margin : 15px  40px  40px  3px;
```

is equivalent to

```
margin-top: 15px;
margin-left : 40px;
margin-right : 40px;
margin-bottom: 30 px;
```

Padding properties

```
padding: 30px
```

is equivalent to

```
padding-top: 30px;
padding-left: 30px;
padding-right: 30px;
padding-bottom: 30px;
```

Number of values	Meaning
one	Assign this value to all four properties (top, right, bottom, left)
two	Assign first value to top & bottom and second value to right & left
three	Assign first value to top, second value to right & left third value to bottom
Four	Assign first-top second-right third-bottom fourth-left

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Image Properties

i) border-image :- used to specify an image as a border around an element.

syntax :-

```
border-image: url("filename") slice repeat;
```

slice ↓ specify how to slice the border image

repeat :- values
* repeat
* round
* stretch

eg:-
```
<style>
#b { border: 10 px solid transparent;
    padding: 15 px;
    border-image: url("flower.jpg") 150 round;
    border-image-width: 4 px;
   }
</style>
<p id="b"> . . . . . </p>
```

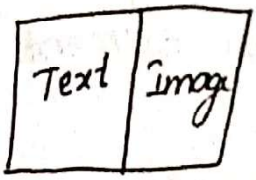
ii) border-radius :- add rounded corners to elements.

syntax: border-radius : values.

eg:-
```
<style>
#b1 { border: 2px solid red;
      border-radius: 25 px;
      padding: 10 px;
    }
</style>
<p id="b1"> . . . . . </p>
```

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iii) To remain image as one partition and text as another partition




```
img
{
float:right;
}
```

Link Properties

Links can be styled differently depending on what state they are in:

1) a:link - a normal, unvisited link
2) a:visited - a link the user has visited
3) a:hover - a link when the user moves over it
4) a:active - a link the moment it is clicked.

eg:

```
<style>
a:hover
{
color:red;
background-color:yellow;
border:2px solid orange;
text-decoration:none;
border-radius:30px;
}
</style>
<body>
<a href=" "> click </a>
</body>
```

o/p:

click

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